

LA POCATIERE, QC, Canada

WMO: 717130

Lat: 47.354N

Lon: 70.029W

Elev: 102

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -4.8	(d) -19.4	(e) 1.9	(f) -7.8	(g) -15.0	(h) 2.4	(i) -3.6	(j) 36.6	(k) 21.8	(l) 33.8	(m) 17.2	(n) 12.1	(o) 220	(p) 0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	83.2	70.2	80.1	68.1	77.5	66.9	72.7	79.8	70.7	77.7	68.8	75.6	14.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	111.5	77.3	68.2	103.8	75.7	66.0	96.1	73.6	36.4	79.8	34.7	77.9	33.1	76.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.7	21.6	-16.2	88.2	4.1	2.8	-19.1	90.2	-21.5	91.8	-23.9	93.4	-26.9	95.4
			-16.6	76.4	4.1	2.6	-19.5	78.2	-21.9	79.7	-24.2	81.2	-27.1	83.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	13.6	16.1	24.8	37.9	51.0	60.3	66.0	64.6	57.1	44.7	33.8	21.1		
	DBStd	20.06	11.38	10.18	9.57	6.96	7.47	7.00	5.53	5.74	7.16	7.19	8.15	10.15		
	HDD50	4909	1128	950	782	369	77	4	0	0	15	199	489	896		
	HDD65	8949	1593	1370	1246	815	439	176	54	77	252	629	937	1361		
	CDD50	1643	0	0	0	4	109	313	496	454	229	35	3	0		
	CDD65	208	0	0	0	0	6	35	85	66	16	0	0	0		
	CDH74	1508	0	0	0	2	87	289	574	436	120	0	0	0		
	CDH80	267	0	0	0	0	18	53	107	73	16	0	0	0		
(14) Wind	WSAvg	9.2	11.8	11.0	10.4	9.2	8.6	7.8	7.6	7.3	8.1	8.9	9.6	10.6		
(15) Precipitation	PrecAvg	37.9	3.1	2.5	2.8	2.7	3.3	3.3	3.5	3.7	3.6	2.8	3.3	3.7		
	PrecMax	56.1	6.8	5.6	6.7	8.5	7.5	5.9	6.7	8.7	6.5	5.5	6.7	8.5		
	PrecMin	30.4	0.8	0.2	0.8	0.6	1.2	1.2	1.5	0.6	0.8	0.9	2.0	0.9		
	PrecStd	6.6	1.4	1.2	1.3	1.5	1.6	1.2	1.4	2.1	1.3	1.1	1.1	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	42.0	50.7	66.1	82.3	85.5	87.1	85.9	82.1	69.9	61.8	45.7		
		MCWB	41.9	38.9	43.9	53.3	63.0	69.1	73.3	71.1	70.4	60.4	56.5	43.6		
	2%	DB	37.9	37.4	43.7	58.5	75.5	80.8	83.0	81.9	77.6	65.0	53.3	39.6		
		MCWB	35.9	34.7	38.7	47.9	60.6	67.2	71.4	70.0	67.8	57.9	48.9	37.1		
	5%	DB	34.3	34.3	40.3	53.4	69.8	77.2	79.7	78.6	73.5	60.6	48.5	36.6		
		MCWB	32.6	32.2	36.2	44.8	57.9	65.7	69.0	68.3	65.2	54.7	44.3	34.2		
	10%	DB	29.9	31.4	37.3	49.0	65.1	73.7	77.0	75.9	70.0	56.3	44.5	34.3		
		MCWB	28.2	29.7	33.9	42.0	54.7	63.3	67.5	66.5	62.6	50.8	41.0	32.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	40.0	44.9	56.9	68.1	73.1	76.7	74.6	72.8	62.8	57.4	43.9		
		MCDB	43.1	41.9	47.8	63.8	78.9	80.9	84.0	82.0	79.5	67.4	61.1	45.5		
	2%	WB	36.3	35.0	39.5	49.2	63.5	69.9	73.2	71.9	69.4	58.9	49.7	37.6		
		MCDB	37.8	36.9	42.5	56.4	71.4	77.3	80.0	79.3	75.7	64.0	52.8	39.4		
	5%	WB	32.7	32.5	36.6	45.8	59.1	67.2	71.1	69.8	66.2	55.3	44.7	34.8		
		MCDB	34.3	34.2	39.7	52.2	67.2	74.4	77.9	76.8	72.1	60.5	47.7	36.2		
	10%	WB	28.3	29.8	34.2	42.8	55.9	64.6	69.1	67.8	63.3	51.4	41.5	32.5		
		MCDB	29.7	31.3	36.8	48.2	64.3	72.0	75.6	74.5	69.4	55.8	44.2	33.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	14.7	14.2	14.8	18.6	18.8	17.8	18.5	17.5	13.4	10.8	11.8		
		MCDBR	15.7	14.6	16.2	23.7	28.1	23.5	21.1	21.2	21.8	19.1	15.9	12.3		
	5% WB	MCWBR	15.4	13.4	12.9	15.9	15.9	13.6	12.2	12.0	13.8	14.0	13.5	11.8		
		MCWBR	15.7	13.7	14.9	21.3	24.7	20.2	19.0	18.9	20.5	18.3	14.9	12.5		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.282	0.320	0.362	0.369	0.393	0.416	0.394	0.372	0.336	0.304	0.279		
	taud		2.342	2.279	2.241	2.298	2.355	2.330	2.289	2.354	2.407	2.494	2.460	2.354		
	Ebn at Noon		262	281	285	282	284	276	267	269	265	259	246	241		
	Edh at Noon		23	30	36	38	38	39	40	36	31	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		432	720	1098	1413	1601	1726	1697	1480	1153	677	381	294		
	RadStd		55	87	84	116	168	114	171	97	88	59	43	28		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page